

retirement spending. Each year, she increases her spending in line with inflation. Although she started spending 4% of her wealth at her start of retirement, her *spending rate* fluctuates up and down depending on asset prices and inflation rates. For individuals retiring over 1926–1976, Bengen found that their wealth was sufficient to last for at least thirty-three years. Increasing the initial withdrawal rate to 6%, wealth would be often exhausted before twenty years, especially for those retiring during the 1960s and early 1970s.

The 4% rule is now generally known as a *fixed proportional spending rule*—spend a constant 4% of your wealth each year. Milevsky (2012) observes that “this 4% rule has taken on the status of a mythical beast in the retirement planning world. (You slay it, or it slays you.)” But the 4% rule is simple, easy to implement, and can be followed regardless of how much wealth an individual has accumulated. Consequently, the 4% rule is very popular. Many financial planners recommend rules similar to the 4% rule—if not 4%, then some other fixed fraction of wealth. It also helps that 4% is very close to the spending rates of endowments and foundations (see chapter 1) and is often used as a long-run average real interest rate (see chapter 11).

The main problem of the fixed 4% rule is that it forces retirees’ consumption to go up and down with the market. That is, individuals are forced to spend more when markets are performing well and cut drastically back when markets crash. Most individuals do not like such volatility in their consumption plans—many of their outflows are fixed in dollar terms, not percentage of wealth terms.<sup>46</sup> Thus retirees tend to prefer regular fixed payments that remain approximately constant in real terms. There is also severe *longevity risk* with the 4% rule. The 4% rule is proportional to wealth, so if you live for a long time your wealth might dwindle to nearly nothing. And 4% of nothing is, well. . . .

Bengen never originally advocated a *fixed* rule of 4%. His was an *initial* spending rule, and he held spending fixed in real terms and adjusted the spending rate over time. The sure-fire way to ensure that your spending never falls, in nominal or real terms, is to follow Dybvig’s ratchet rule. In Bengen’s formulation, there is always risk that an investor will outlive his savings: even if in Bengen’s original sample the retirement nest egg lasted for at least thirty years, pity the poor seniors who live past one hundred who will eventually have nothing.

Finance theory also says that the spending rate (consumption) should be determined simultaneously with the portfolio choice. If you’re holding a risky portfolio, you probably want to spend less, and vice versa. This applies to both retirement and to how we save and consume getting to retirement. The 4% rule says nothing about optimal portfolio choice. The best financial planners do not use the 4% rule and instead craft a retirement plan that takes into account longevity risk, health

<sup>46</sup> Harvard University experienced some problems in meeting liabilities with its fixed spending rule from its endowment. See “Liquidating Harvard,” Columbia CaseWorks ID #100312.